Digital Communication IC Design

Homework 4

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Development Environment

Language	MATLAB R2025b, SystemVerilog
Simulator	Synopsys VCS S-2021.09, Verdi T-2022.06
AI Tool	Claude Opus 5.5

Due Date: 2026/11/18

Contents

1 Setup and Code Organization **2**

2 Q1 — Receive Beamforming with a ULA **3**

2.1 (a) AoA, Path Gains and Channel Vector . . . 3

2.2 (b) Beam Pattern of $\mathbf{c}(m_1)$ . . . 3

2.3 (c) Combiner Output versus m . . . 5

2.4 (d) Nominal Pointing Angle of the Selected Beam . . . 5

2.5 (e) Which Path Does the Selected Beam Follow? . . . 5

3 Q2 — Symbol Timing by Delay-and-Correlate **7**

3.1 (a) $\Phi_{DC}(m)$ with $R = 64, L = 64$ . . . 7

3.2 (b) Detected Timing and Plateau in the First Symbol . . . 8

3.3 (c) $\Phi_{DC}(m)$ and $\Phi_{NM}(m)$ with $R = 32, L = 128$ . . . 9

3.4 (d) $\Phi_{NM}(m)$ at SNR = 12 dB . . . 11

4 Q3 — Hardware Symbol Boundary Detector **12**

4.1 (a) Threshold η . . . 12

4.2 (b) Block Diagram . . . 14

4.3 (c) Word Lengths . . . 15

4.4 (d) HDL Implementation and Simulation . . . 16

4.4.1 Implementation . . . 16

4.4.2 Latency of the Local Peak Searcher . . . 16

4.4.3 Behavioral Simulation . . . 17

4.4.4 Comparison with the Simulation . . . 18

5 AI Tool Disclosure **20**

Setup and Code Organization

My student ID is M11407439. Its last digit, 9, is m_1 in Q1(b), and the whole number 11407439 is the random seed for the angles and path gains of Q1 and for the noise of Q2(d) and Q3. The MATLAB scripts are in `01_src` and the hardware design is in `02_hdl`.

Each question has one MATLAB script (`q1.m` to `q3.m`); the RTL is in `02_hdl/01_RTL` and the testbench in `02_hdl/00_TESTBED`. All numbers in this report come from the console output of the scripts and of the VCS simulation.

Q1 — Receive Beamforming with a ULA

Let

$$\mathbf{h} = \frac{1}{\sqrt{L}} \sum_{l=1}^{L} \alpha_l \, \mathbf{a}^{ULA}(\theta_{r,l}), \qquad \mathbf{a}^{ULA}(\theta) = \frac{1}{\sqrt{N_r}} \begin{bmatrix} 1 & e^{-jkd\sin(\theta)} & \cdots & e^{-j(N-1)kd\sin(\theta)} \end{bmatrix}^T.$$

Assume $kd = \pi$ and 64 ($= N_r$) receiving antenna elements are used in the uniform linear antenna array. The transmitter has one antenna. Given $L = 2$.

2.1 (a) AoA, Path Gains and Channel Vector

Please generate your own $\theta_{r,1}$ and $\theta_{r,2}$ such that $-\pi/3 < \theta_{r,1}, \theta_{r,2} < \pi/3$. Report them in both radians and degrees. Set $\theta_{t,1} = \pi/7$. Please generate your channel matrix $\mathbf{h}$ with AoA $\theta_{r,1}$ and $\theta_{r,2}$, which is a vector of dimension 64×1. Let α_1 and α_2 be the complex channel coefficients with $|\alpha_1| > 0.5 > |\alpha_2|$. Please generate them randomly and report their magnitudes and phases. (5%)

With `rng(11407439)`, I draw both angles from the uniform distribution on $(-\pi/3, \pi/3)$. I draw the magnitudes as $|\alpha_1| = 0.5 + 0.5\,U$ and $|\alpha_2| = 0.5\,U$ with $U \sim \mathcal{U}(0,1)$, so $|\alpha_1| > 0.5 > |\alpha_2|$ holds by construction. Both phases are uniform in $(-\pi, \pi)$.

Table 2.1: Angles of Arrival and Path Gains

Path l	$\theta_{r,l}$ (rad)	$\theta_{r,l}$ (deg)	$\lvert\alpha_l\rvert$	$\angle\alpha_l$ (rad / deg)
1	−0.883475	−50.6194°	0.732884	+0.372299 / +21.3311°
2	−0.775469	−44.4311°	0.431265	+2.194091 / +125.7121°

That is, $\alpha_1 = 0.682677 + 0.266592\,j$ and $\alpha_2 = -0.251735 + 0.350170\,j$. The channel vector is

$$\mathbf{h} = \frac{1}{\sqrt{2}} \left(\alpha_1 \, \mathbf{a}^{ULA}(\theta_{r,1}) + \alpha_2 \, \mathbf{a}^{ULA}(\theta_{r,2}) \right), \qquad [\mathbf{a}^{ULA}(\theta)]_n = \frac{1}{8} \, e^{-j\pi n \sin\theta}, \quad n = 0, \ldots, 63, \tag{2.1}$$

a 64×1 vector with $\|\mathbf{h}\|^2 = 0.385058$. With a single transmit antenna the transmit array response is 1, so $\theta_{t,1} = \pi/7$ (25.71°) does not appear in $\mathbf{h}$.

2.2 (b) Beam Pattern of $\mathbf{c}(m_1)$

Now let's check the effect of a receiver performing beamforming. Given that

$$\mathbf{c}(m) = \frac{1}{\sqrt{N_r}} \begin{bmatrix} 1 & e^{-j2\pi m/N} & \cdots & e^{-j2\pi(N-1)m/N} \end{bmatrix}^T$$

is used as the phase shift for combiner at the receiver and $m = -32, -31, \ldots, 31$. First, use your last digit of student ID as m_1. Then, draw the beam pattern associated with m_1. Namely, fix $\mathbf{c}(m_1)$ and scan $|\mathbf{c}(m_1)^H \mathbf{a}^{ULA}(\theta)|$ for $-\pi/2 \le \theta \le \pi/2$ in polar coordinate. (10%)

The last digit of M11407439 is 9, so $m_1 = 9$. With $N = N_r = 64$ and $kd = \pi$,

$$\left|\mathbf{c}(m)^H \mathbf{a}^{ULA}(\theta)\right| = \left|\frac{\sin(N\psi/2)}{N\sin(\psi/2)}\right|, \qquad \psi = \frac{2\pi m}{N} - \pi\sin\theta. \tag{2.2}$$

The pattern equals 1 at $\psi = 0$, i.e. $\sin\theta = 2m/N$, which gives the nominal pointing angle

$$\varphi(m) = \sin^{-1}\left(\frac{2m}{N}\right), \qquad \varphi(m_1) = \sin^{-1}\left(\frac{18}{64}\right) = 0.285096 \text{ rad} = 16.3348^\circ. \tag{2.3}$$

The scan over 4001 angles peaks at 16.335° with value 1.000000.

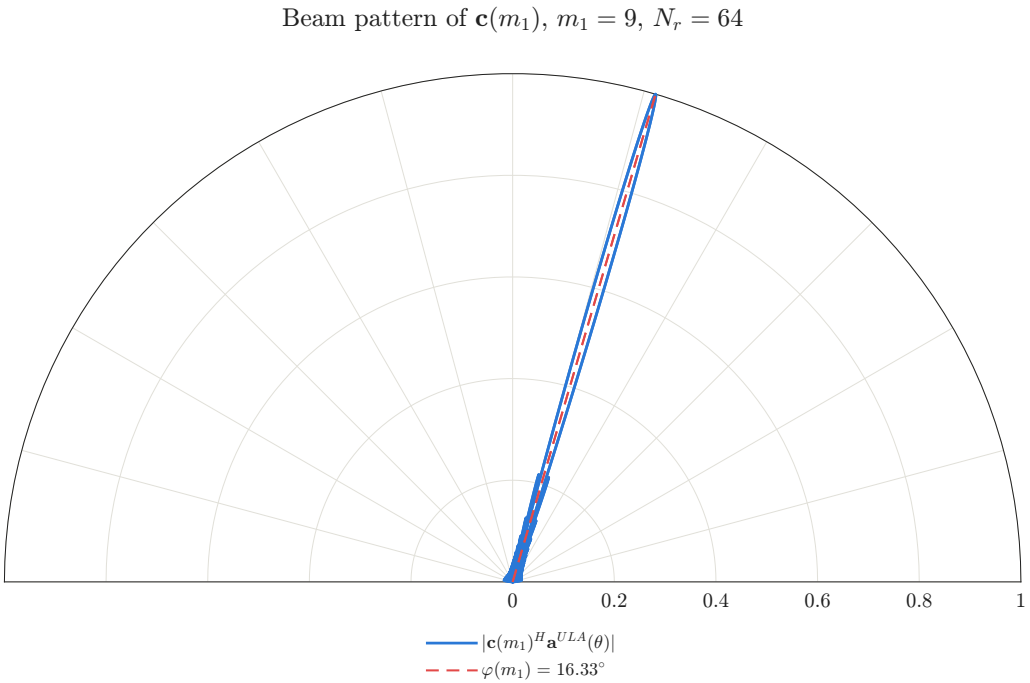


Figure 2.1: Beam Pattern $|\mathbf{c}(m_1)^H \mathbf{a}^{ULA}(\theta)|$ for $m_1 = 9$ in Polar Coordinates

2.3 (c) Combiner Output versus m

Please check the magnitude of the combination result, $|\mathbf{c}(m)^H\mathbf{h}|$ versus m. Draw the figure with x-axis indicating the value of m and y-axis is the magnitude of $|\mathbf{c}(m)^H\mathbf{h}|$ for $m = -32, -31, \ldots, 31$. (10%)

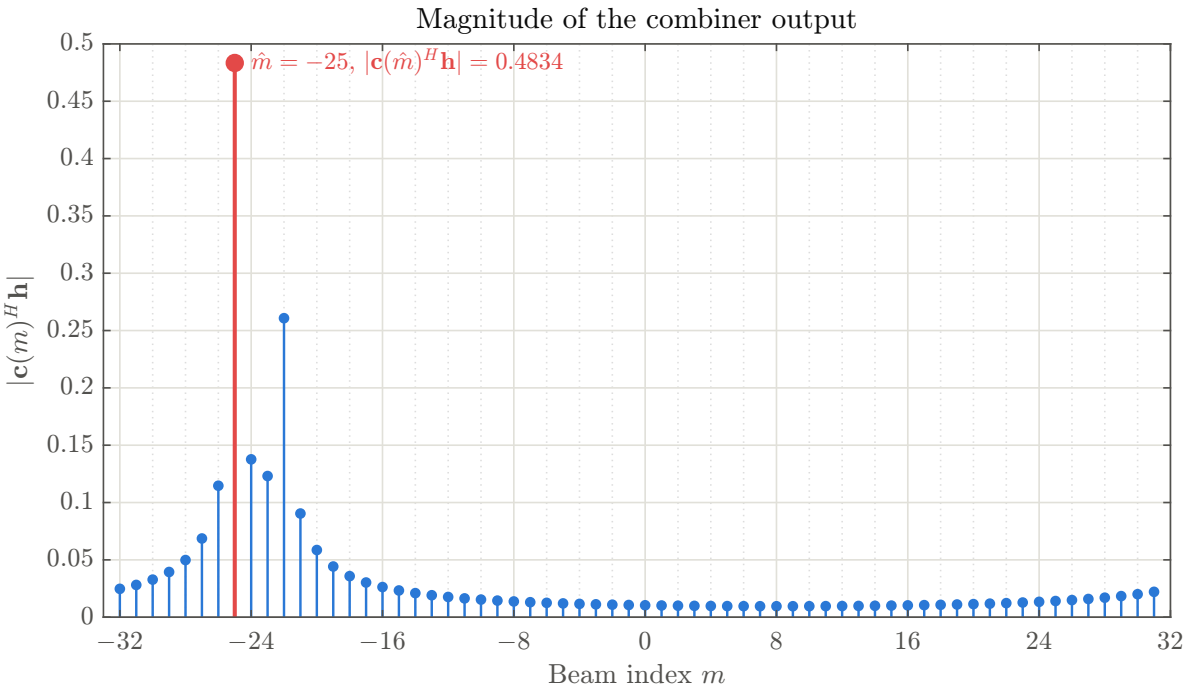


Figure 2.2: Magnitude of the Combiner Output $|\mathbf{c}(m)^H\mathbf{h}|$ for $m = -32 \sim 31$

The output is large only for $m = -25 \sim -21$, where the two paths arrive: 0.4834 at $m = -25$ and a second peak of 0.2608 at $m = -22$. On the beam-index axis, path 1 lies at $N \sin\theta_{r,1}/2 = -24.73$ and path 2 at -22.40. Neither is an integer, so some power leaks into the neighboring beams.

2.4 (d) Nominal Pointing Angle of the Selected Beam

If $\hat{m} = \arg\max_m |\mathbf{c}(m)^H\mathbf{H}_1|$, then calculate the nominal pointing angle $\varphi(\hat{m})$ associated with the beamforming vector $\mathbf{c}(\hat{m})$ (5%). Please check the differences of $|\varphi(\hat{m}) - \theta_{r,1}|$ and $|\varphi(\hat{m}) - \theta_{r,2}|$. (5%)

Here $\mathbf{H}_1$ is the vector $\mathbf{h}$ from (a). From (c), $\hat{m} = -25$ and

$$\varphi(\hat{m}) = \sin^{-1}\left(\frac{2\times(-25)}{64}\right) = -0.896666 \text{ rad} = -51.3752^\circ, \tag{2.4}$$

$$|\varphi(\hat{m}) - \theta_{r,1}| = 0.013191 \text{ rad} = 0.7558^\circ, \tag{2.5}$$

$$|\varphi(\hat{m}) - \theta_{r,2}| = 0.121197 \text{ rad} = 6.9441^\circ. \tag{2.6}$$

2.5 (e) Which Path Does the Selected Beam Follow?

Explain whether the selected beam is closer to the stronger path, the weaker path, or neither. Support the explanation using the path magnitudes, path phases, angular separation, and the values. (5%)

The selected beam is closer to the stronger path, path 1. Table 2.2 splits $\mathbf{c}(\hat{m})^H\mathbf{h}$ into the contribution of each path, $\frac{\alpha_l}{\sqrt{2}}\,\mathbf{c}(\hat{m})^H\mathbf{a}^{ULA}(\theta_{r,l})$.

Table 2.2: Contribution of Each Path to the Selected Beam $\hat{m} = -25$

Path	$\lvert\alpha_l\rvert$	$\angle\alpha_l$	$N \sin\theta_{r,l}/2$	$\lvert\mathbf{c}(\hat{m})^H\mathbf{a}(\theta_{r,l})\rvert$	|contribution|	∠contribution
1	0.7329	+21.33°	−24.73	0.8879	0.4601	−25.74°
2	0.4313	+125.71°	−22.40	0.1170	0.0357	+25.31°

Path 1 is the stronger one ($|\alpha_1| = 0.733$ vs. $|\alpha_2| = 0.431$). The two paths are 6.19° apart, which is 2.33 beam spacings and wider than the 5.75° null-to-null width of beam −25, so each path has its own best beam: path 1 is 0.27 beam from $m = -25$ and path 2 is 0.40 beam from $m = -22$. In beam −25, path 1 keeps a gain of 0.888, while path 2 only leaks in through a sidelobe (0.117), so its contribution is 0.0357 against 0.4601. The two contributions are 51° apart in phase, so their sum, 0.4834, exceeds 0.4601 by 0.0233. A term of 0.036 cannot move the maximum, whatever its phase. The best beam for path 2, $m = -22$, reaches 0.2608.

Q2 — Symbol Timing by Delay-and-Correlate

Please download "hw4.mat" from the NTUCool. Use "load hw4.mat" to retrieve the original data. It contains one variable "OFDMTx", which are the complex time-domain discrete signals generated by OFDM baseband transmitter. There are 32 samples in cyclic prefix ($N_g = 32$) and the FFT size is 128 ($N = 128$). The first OFDM symbol is a preamble which has identical two repetitions as below, and then three normal OFDM symbols follows the preamble. Assume the received signal z_m equal to the transmitted signal. If $m \le 0$, $z_m = 0$.

`OFDMTx` has 640 samples, four symbols of $N_g + N = 160$ samples. The preamble, CP included, is periodic with period $N/4 = 32$: the script finds samples 33–64, 65–96, 97–128 and 129–160 identical, and the CP equals 129–160. Indices start from 1, so symbol s ends at sample $160s$.

3.1 (a) $\Phi_{DC}(m)$ with $R = 64$, $L = 64$

Please draw magnitude of the delay and correlate result $\Phi_{DC}(m)$ by setting $R = 64$ and $L = 64$, for $m = 1, \ldots, 640$ where

$$\Phi_{DC}(m) = \left| \sum_{r=0}^{R-1} z_{m-r}\, z^*_{m-r-L} \right|. \quad (10\%)$$

(Mark the X and Y value of the peak in the figure. The first index in the figure is 1.)

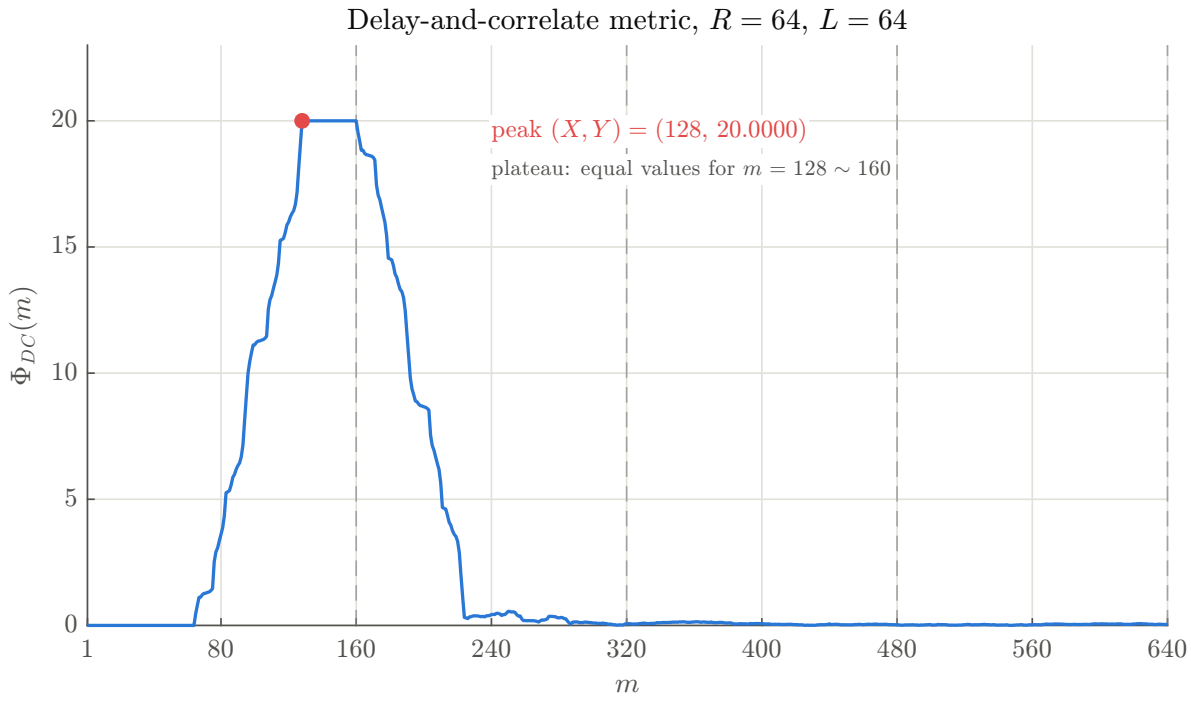


Figure 3.1: $\Phi_{DC}(m)$ for $R = L = 64$, $m = 1 \sim 640$ (dashed lines: symbol boundaries)

The peak is at $(X, Y) = (128,\ 20.0000)$. Φ_{DC} is 0 for $m \le 64$ because $z_{m-r-64} = 0$ there, rises while the

delayed window moves into the preamble, stays at 20 for $m = 128 \sim 160$, and drops once the window reaches the data symbols.

3.2 (b) Detected Timing and Plateau in the First Symbol

Consider only the first symbol for $1 \le m \le 160$. If a decision is made according to $\hat{m}_{DC} = \arg\max \Phi_{DC}(m)$, what's the detected timing in the first symbol? (5%) Whether a plateau will be seen or not in the waveform of $\Phi_{DC}(m)$ using the setting in Q2(a)? Why? (5%)

The detected timing is $\hat{m}_{DC} = 128$ with $\Phi_{DC} = 20$, the first sample of the maximum: all 33 samples from 128 to 160 have the same value, and `max` returns the first of them. The preamble ends at 160.

Φ_{DC} shows a plateau. $\Phi_{DC}(m)$ uses the 128 samples $z_{m-127}, \ldots, z_m$. If all of them lie inside the periodic preamble ($m - 127 \ge 1$ and $m \le 160$), every product is $z_{m-r} z^*_{m-r-64} = |z_{m-r}|^2$, because the lag 64 is a multiple of the period 32. Then

$$\Phi_{DC}(m) = \sum_{r=0}^{63} |z_{m-r}|^2 = 20, \qquad 128 \le m \le 160, \tag{3.1}$$

and the value stays constant because a 64-sample window covers two full periods. The plateau is $160 - (R + L) + 1 = 33$ samples long, which is positive because the repeated part (160 samples, CP included) is longer than $R + L = 128$. At $m = 161$ the first data sample enters and the value drops to 19.56.

3.3 (c) $\Phi_{DC}(m)$ and $\Phi_{NM}(m)$ with $R = 32$, $L = 128$

Discard the first symbol. Now, the signal length is 480 sample. Please draw the result $\Phi_{NM}(m)$ and $\Phi_{DC}(m)$ by setting $R = 32$ and $L = 128$ for $m = 1, \ldots, 480$. If a decision is made according to $\hat{m}_{DC} = \arg\max \Phi_{DC}(m)$ and $\hat{m}_{NM} = \arg\max \Phi_{NM}(m)$ every symbol, what are the results? Please explain the decision difference between $\Phi_{NM}(m)$ and $\Phi_{DC}(m)$. (15%)

$$\Phi_{NM}(m) = \frac{\left|\sum_{r=0}^{R-1} z_{m-r}\, z_{m-r-L}^{*}\right|^2}{\left(\sum_{r=0}^{R-1} |z_{m-r}|^2\right)^2}$$

With $R = N_g$ and $L = N$, the window compares the last 32 samples of a symbol with its CP, so the exact match is at $m = 160, 320, 480$. I take "every symbol" as the blocks $m = 1 \sim 160$, $161 \sim 320$ and $321 \sim 480$.

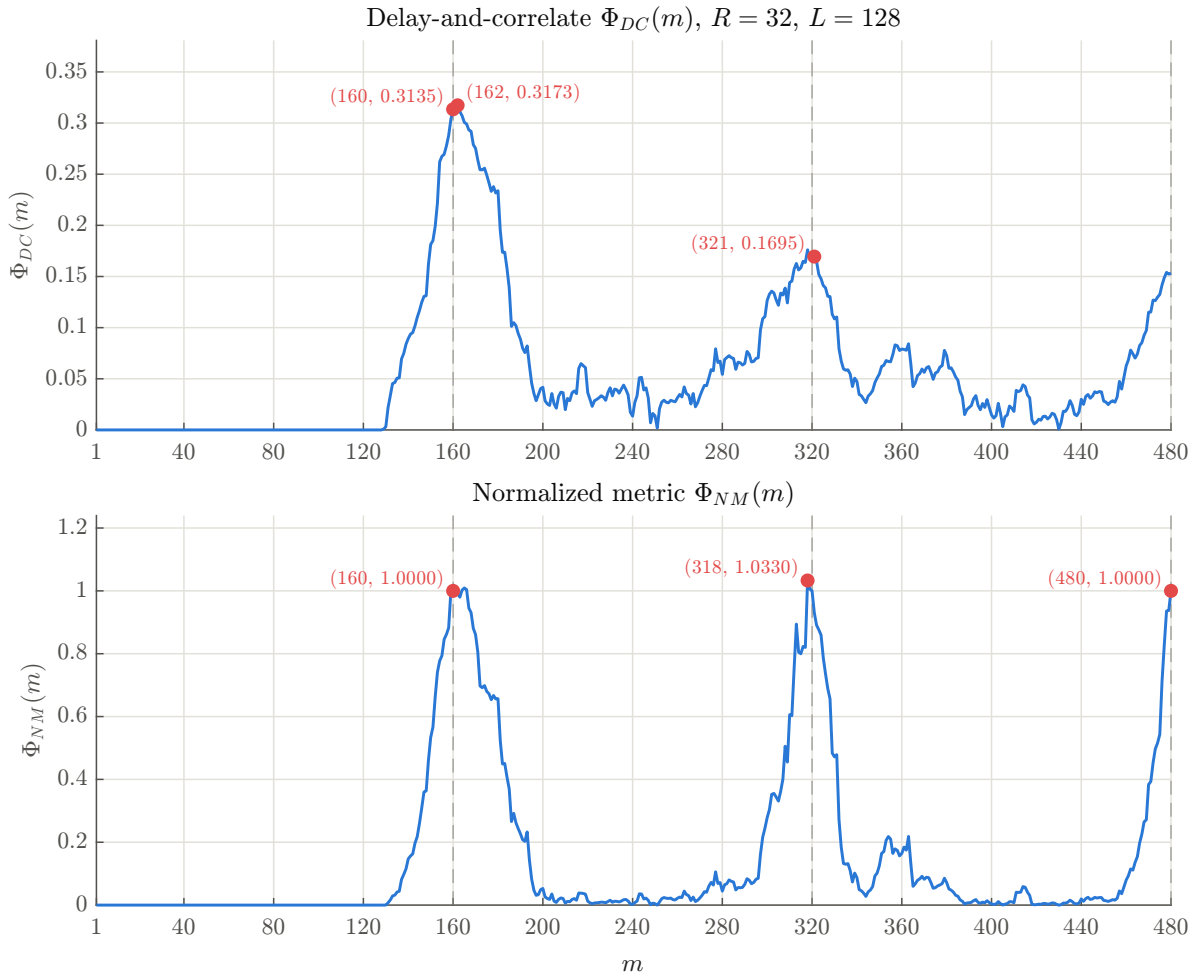


Figure 3.2: $\Phi_{DC}(m)$ (top) and $\Phi_{NM}(m)$ (bottom) for $R = 32$, $L = 128$, with the Maximum of Every Symbol

Table 3.1: Decision in Every Symbol (Noiseless)

Symbol	Block	Symbol end	$\hat{m}_{DC}$ (Φ_{DC})	$\hat{m}_{NM}$ (Φ_{NM})	$\sum_r |z_{m-r}|^2$ at symbol end
1	$1 \sim 160$	160	160 (0.3135)	160 (1.0000)	0.3135
2	$161 \sim 320$	320	162 (0.3173)	318 (1.0330)	0.1680
3	$321 \sim 480$	480	321 (0.1695)	480 (1.0000)	0.1532

At the symbol end every product is $|z|^2$, so Φ_{DC} there equals the energy of the window, and that energy differs between symbols (0.3135, 0.1680, 0.1532). Symbol 1 is the strongest. The falling edge of its lobe is still 0.3173 at $m = 162$, above the true peak of symbol 2 (0.1680), so block 2 picks 162, which belongs to the previous boundary. Block 3 fails the same way: 321 is the tail of the symbol-2 lobe (0.1695) and beats the symbol-3 peak (0.1532), so Φ_{DC} finds symbol 1 alone.

Φ_{NM} divides by the squared window energy, so an exact CP match gives 1 for every symbol, whatever its power. It finds 160 and 480. In block 2 it picks 318 with $\Phi_{NM} = 1.033$, two samples early. Φ_{NM} can exceed 1 because the denominator only uses the current window: by Cauchy–Schwarz, $\Phi_{NM}(m) \le \sum_r |z_{m-r-L}|^2 / \sum_r |z_{m-r}|^2$, and this ratio is above 1 whenever the delayed window carries more energy. The decisions therefore differ because Φ_{DC} depends on the symbol power and Φ_{NM} does not.

3.4 (d) $\Phi_{NM}(m)$ at SNR = 12 dB

Now, add noise by yourself. The SNR is 12dB. Please use normalized metric $\Phi_{NM}(m)$ to perform symbol timing detection again. Report the detected result $\hat{m}_{NM} = \arg\max \Phi_{NM}(m)$ every symbol and compare the results to the answer in Q2(c). Please explain the difference if any. (10%)

I add circular complex Gaussian noise to the 480-sample record with $\sigma^2 = P_s/10^{1.2}$, where $P_s = \overline{|z_m|^2} = 0.007585$ is the record power, so $\sigma^2 = 4.79 \times 10^{-4}$ (seed 11407439). The SNR measured on this realization is 12.07 dB.

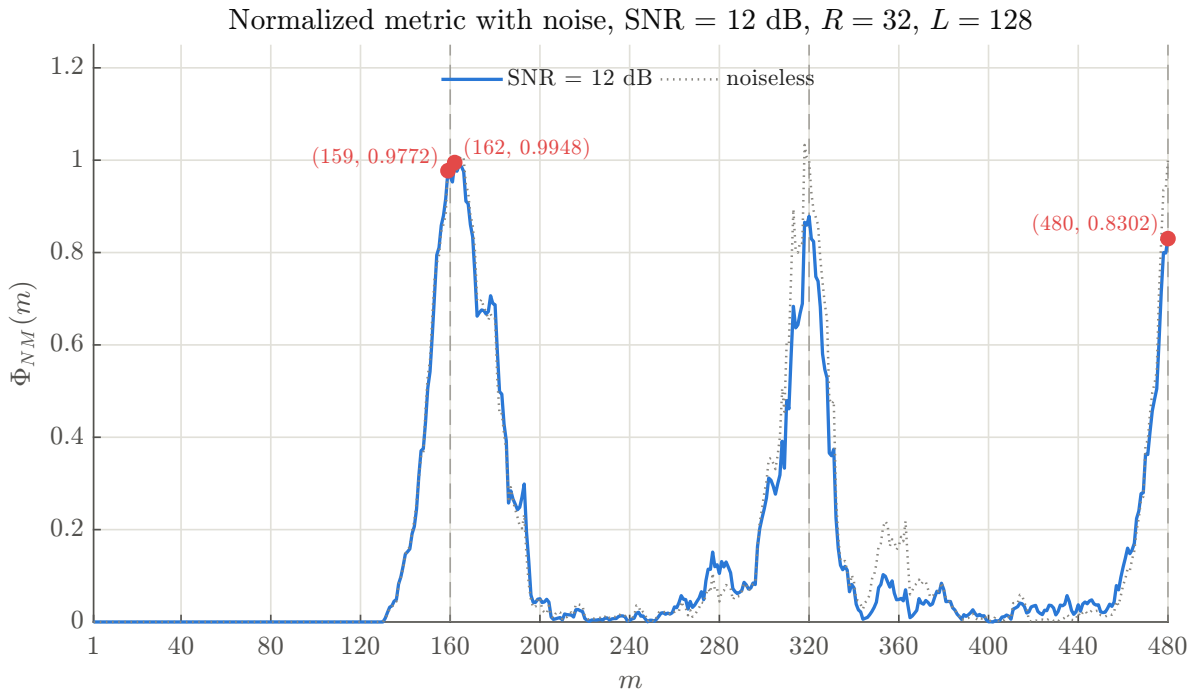


Figure 3.3: $\Phi_{NM}(m)$ at SNR = 12 dB (blue) and Noiseless (dotted), with the Maximum of Every Symbol

Table 3.2: Decision of Φ_{NM} in Every Symbol, SNR = 12 dB versus Noiseless

Symbol	$\hat{m}_{NM}$ at 12 dB (Φ_{NM})	$\hat{m}_{NM}$ in Q2(c)	Φ_{NM} at symbol end, 12 dB	Local SNR
1	159 (0.9772)	160	0.9643	13.1 dB
2	162 (0.9948)	318	0.8784	10.4 dB
3	480 (0.8302)	480	0.8302	10.0 dB

Symbol 3 is unchanged, symbol 1 moves one sample earlier, and symbol 2 is now wrong. With noise the CP and its copy no longer match, and the peak drops to about $\left(S/(S+\sigma^2)\right)^2$, where S is the signal power in the window. The noise power is the same everywhere, but the CP windows carry different power, so the local SNR is 13.1, 10.4 and 10.0 dB and the expected peaks are about 0.91, 0.84 and 0.83. The measured values at the symbol ends (0.964, 0.878, 0.830) follow the same order.

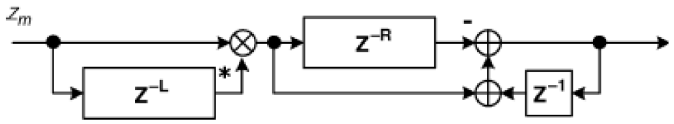
The lower peak breaks symbol 2. Its value drops to 0.878, while the falling edge of the symbol-1 lobe at $m = 162$, which has the highest local SNR, stays at 0.995. Both lie in block 2, so the argmax returns 162, the same mistake Φ_{DC} made in Q2(c). Each lobe is $2R$ wide and centered on a block edge, so half of each lobe falls into the next block. Symbol 1 moves to 159 because the top of its lobe is almost flat (0.977 vs. 0.964) and the noise picks the winner.

Q3 — Hardware Symbol Boundary Detector

Now, we aim to implement $\Phi_{NM}(m)$. A constant threshold η is set to activate the peak searcher. It means the local maximum will be searched only when the correlation output is larger than η, namely

$$\Phi_{NM}(m) = \frac{\left|\sum_{r=0}^{R-1} z_{m-r}\, z_{m-r-L}^{*}\right|^2}{\left(\sum_{r=0}^{R-1} |z_{m-r}|^2\right)^2} > \eta.$$

The delay correlation is implemented for the numerator as in the following.



The metric above follows the TA's correction, which replaced the original $\Phi_{NM}(m) = \left|\sum_{r=0}^{R-1} z_{m-r} z_{m-r-L}^{*}\right|$ with the normalized metric of Q2(c). The hardware uses the setting of Q2(c), $R = 32$ and $L = 128$, on the 480-sample record, with one input sample per clock cycle.

4.1 (a) Threshold η

Decide your own setting of η from the given signals and the signals with SNR equal to 12dB. Please report the setting and explain the reason. (5%)

I set $\eta = 0.52$, stored as the U2.14 code 8520 (`0x2148`).

Around each boundary Φ_{NM} forms a lobe $2R$ samples wide ($|m - 160k| < R$, shaded in the figure below). The top of each lobe must pass η, and the values outside the lobes must stay below it, otherwise the peak searcher starts on data samples.

Table 4.1: Φ_{NM} inside and outside the Lobes

Signal	Lobe peaks (symbol 1, 2, 3)	Max. outside the lobes
Given signal (noiseless)	1.0092, 1.0330, 1.0000	0.2328
12 dB, record of Q2(d)	0.9948, 0.8784, 0.8302	0.2987
12 dB, 2000 runs: min. peak / max. outside	0.6913, 0.6220, 0.5669	0.4831
12 dB, 2000 runs: 0.1% / 99.9% quantile	0.7141, 0.6301, 0.5796	0.4614

For the given signal any η between 0.24 and 1.0 works, and the noisy record of Q2(d) still leaves a gap from 0.30 to 0.83. One noisy record says little about other noise realizations, so I also ran 2000 of them at 12 dB. The lobe peaks then fall to 0.567 and the largest value outside the lobes rises to 0.483. I place η halfway between the 99.9% quantile outside the lobes and the 0.1% quantile of the peaks:

$$\eta = \frac{0.461 + 0.580}{2} \approx 0.52. \tag{4.1}$$

Over the 2000 runs this leaves margins of 0.047 below the lowest peak and 0.037 above the highest outside value, and each run gives three detections with the peak searcher of (d).

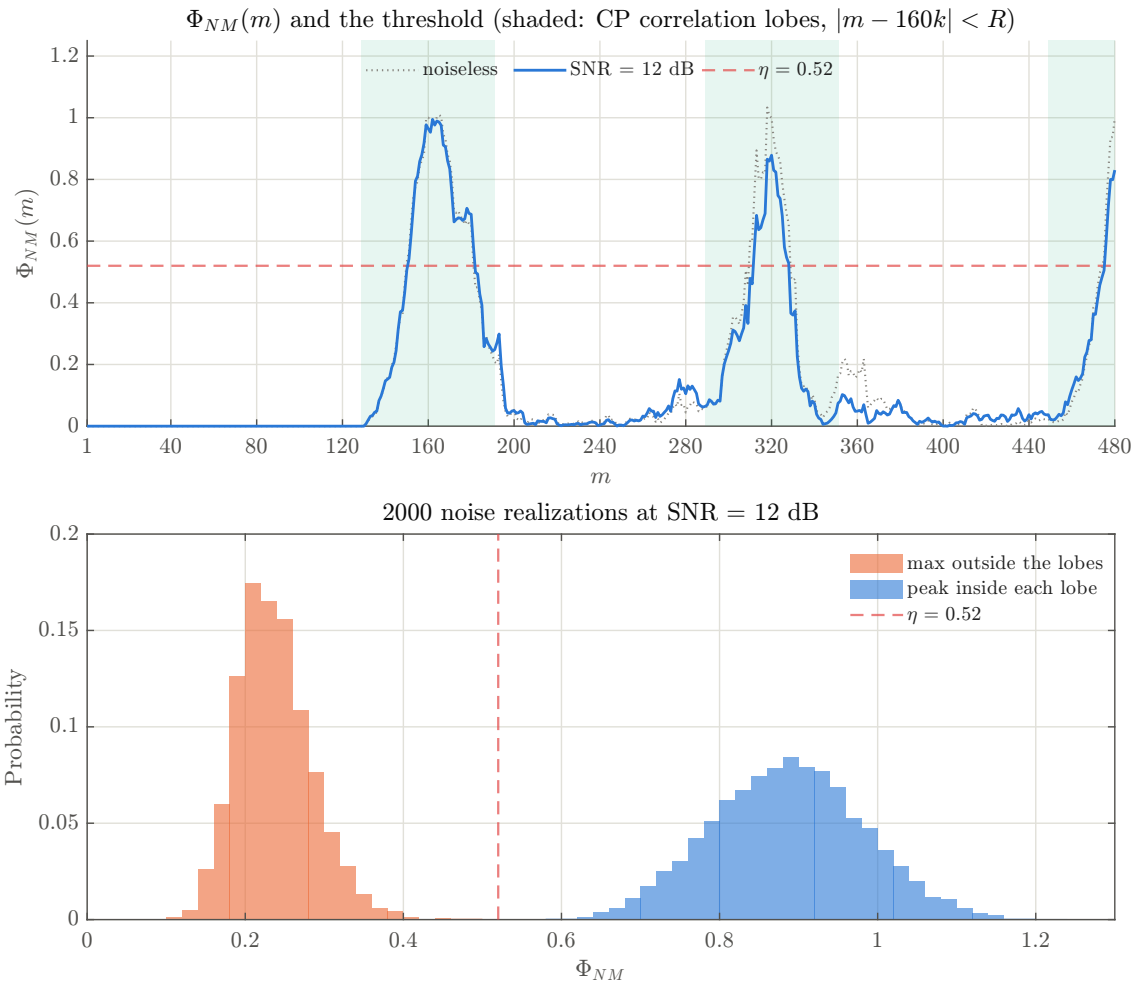
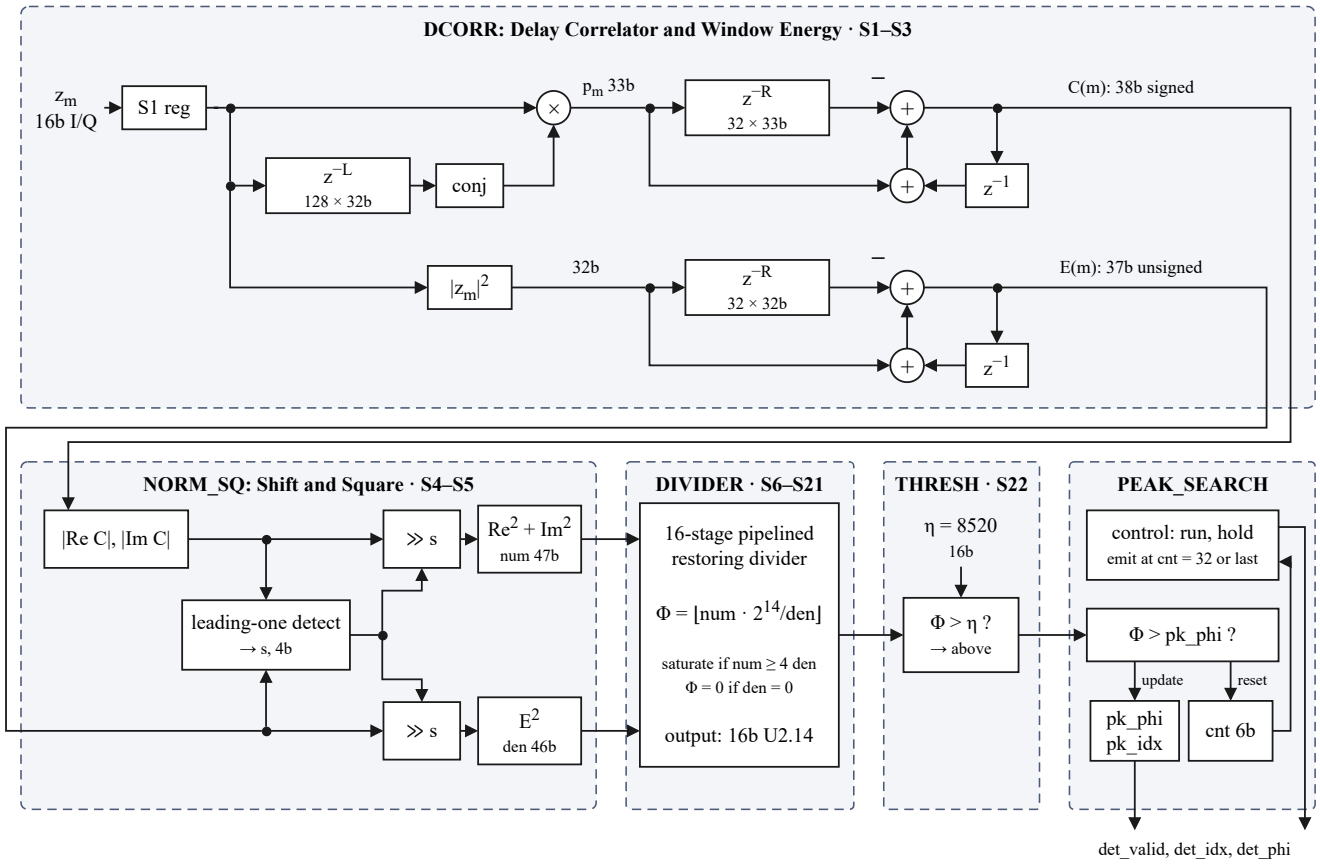


Figure 4.1: $\Phi_{NM}(m)$ with $\eta = 0.52$ (top) and the Distribution of the Lobe Peaks and of the Largest Value outside the Lobes over 2000 Noise Realizations (bottom)

4.2 (b) Block Diagram

Assume that the complex input signal enters one by one sequentially. Draw the block diagram of your design for the threshold-judgement module, divider, and the peak searcher together with the delay correlator. Connect all these modules together in one figure. (10%)



Every stage also carries the 10-bit sample index m, valid and last. z_m to Φ: 22 cycles; peak decision: W + 1 = 33 cycles after the peak.

Figure 4.2: Block Diagram of the Symbol Boundary Detector

One sample z_m (16-bit I and Q) enters per clock cycle. `DCORR` follows the recursive structure of the question, $C(m) = C(m-1) + p_m - p_{m-R}$ with $p_m = z_m z^*_{m-L}$, and a second accumulator of the same form builds the energy $E(m)$. The sample index m travels down the pipeline with the data, so each output carries the index of its own sample.

4.3 (c) Word Lengths

Please quantize your input signal z_m by 16 bits. To avoid overflow, please explain the wordlength setting of your accumulator and your divider. (5%)

After the preamble, the largest $|\mathrm{Re}\, z_m|$ or $|\mathrm{Im}\, z_m|$ is 0.223, noise included. I quantize I and Q to 16-bit two's complement with 16 fraction bits, $z_q = \mathrm{round}(z \cdot 2^{16})$, which covers $[-0.5, 0.5)$ with an LSB of 2^{-16}. The scale factor cancels in the ratio Φ_{NM}, so it sets the resolution of the metric.

The word lengths below follow from the worst-case input $|\mathrm{Re}\, z|, |\mathrm{Im}\, z| \le 2^{15}$ (LSB units).

Table 4.2: Word Lengths

Signal	Bits	Reason
z_m (I, Q)	16 signed	given
$\mathrm{Re/Im}\{z_m z^*_{m-L}\}$	33 signed	sum of two 16×16 products, $\lvert ac + bd\rvert \le 2^{31}$
$\lvert z_m\rvert^2$	32 unsigned	$a^2 + b^2 \le 2^{31}$
Accumulator $C(m)$	38 signed	$R = 32$ terms add $\log_2 R = 5$ bits
Accumulator $E(m)$	37 unsigned	same, from 32 bits
After the common shift	23	top 23 bits of the largest of $\lvert\mathrm{Re}\, C\rvert$, $\lvert\mathrm{Im}\, C\rvert$, E
Numerator $\lvert C\rvert^2$	47 unsigned	$2 \times 23 + 1$
Denominator E^2	46 unsigned	2×23
Dividend num $\cdot\, 2^{14}$	61	also the partial remainder of each divider stage
Quotient Φ	16 unsigned	U2.14, range $[0, 4)$, LSB 2^{-14}

Accumulator. With 38 and 37 bits the accumulators hold the full sum of 32 worst-case terms, so they cannot overflow. In the recursive form $C(m) = C(m-1) + p_m - p_{m-R}$ the register equals the true window sum after each update.

Divider. Squaring the full sums would need 73-bit operands. Scaling C and E by the same factor leaves Φ_{NM} unchanged, so `NORM_SQ` shifts all three values right by $s = \max(0, \mathrm{msb} - 22)$, where msb is the leading-one position of the largest one. The divider then works on 47/46-bit operands. The quotient needs two integer bits because Φ_{NM} can exceed 1 (1.033 in Q2(c)). It saturates to $4 - 2^{-14}$ if num ≥ 4 den and outputs 0 if den $= 0$, so it cannot overflow either. On the given data the largest shift is 8, and the hardware metric differs from the floating-point Φ_{NM} by at most 7.2×10^{-5} (noiseless) and 9.3×10^{-5} (12 dB), i.e. at most 1.5 LSB.

4.4 (d) HDL Implementation and Simulation

Implement your design by hardware description language (HDL) (20%). Explain the latency of your local peak searcher (5%). Show the behavior simulation results (timing diagram) of your symbol boundary detector (10%). Please compare your implementation results with your simulation result in Q3(c) and provide your comment. (5%)

4.4.1 Implementation

I wrote the design in SystemVerilog (`02_hdl/01_RTL`) with the modules of Figure 4.2. The pipeline from input to metric is:

Stage	Module	Operation
S1	`DCORR`	input register, sample index counter
S2	`DCORR`	z^{-L} delay line, $z_m z^*_{m-L}$ and $\lvert z_m \rvert^2$
S3	`DCORR`	z^{-R} delay line, recursive sums $C(m)$, $E(m)$
S4	`NORM_SQ`	leading-one detection and common shift
S5	`NORM_SQ`	squares $\lvert C \rvert^2$, E^2
S6 – S21	`DIVIDER`	16 restoring stages, one quotient bit each
S22	`THRESH`	$\Phi > \eta$

$\Phi(m)$ leaves `THRESH` 22 cycles after z_m enters, and one new value comes out every cycle.

The peak searcher processes one metric sample per cycle:

(1) When idle and $\Phi(m) > \eta$, it starts a search with candidate $(m, \Phi(m))$ and counter 0.

(2) While searching, a value above η that is larger than the candidate replaces it and resets the counter. Any other sample increases the counter.

(3) When the counter reaches $W = 32$, it outputs the candidate (`det_valid`, `det_idx`) and becomes idle. A new search needs $\Phi \leq \eta$ first, so each lobe gives one detection.

(4) At the last sample of the record (`in_last`), a running search is output at once.

I use $W = R = 32$ because a lobe extends R samples on each side of its peak; after 32 samples without a larger value the search has left the lobe.

4.4.2 Latency of the Local Peak Searcher

The searcher can call a value the local maximum only after it sees the following samples. The searcher waits $W = 32$ metric samples and then registers the result, so `det_valid` rises

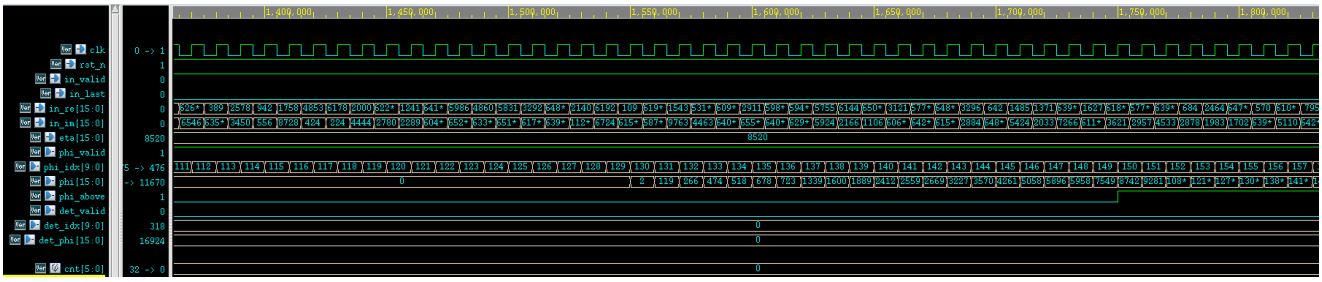
$$\underbrace{W}_{\text{32 samples}} + \underbrace{1}_{\text{output register}} = 33 \text{ cycles} \tag{4.2}$$

after $\Phi(\hat{m})$ reaches the searcher, or $22 + 33 = 55$ cycles after $z_{\hat{m}}$ enters the design. The simulation measures 55 cycles for the boundaries 165 and 318 of the noiseless case and 162 and 320 of the 12 dB case. For the last boundary, $m = 480$, the flush rule outputs the candidate one cycle after the last sample. The waveforms in the next subsection show these cases for the noiseless record.

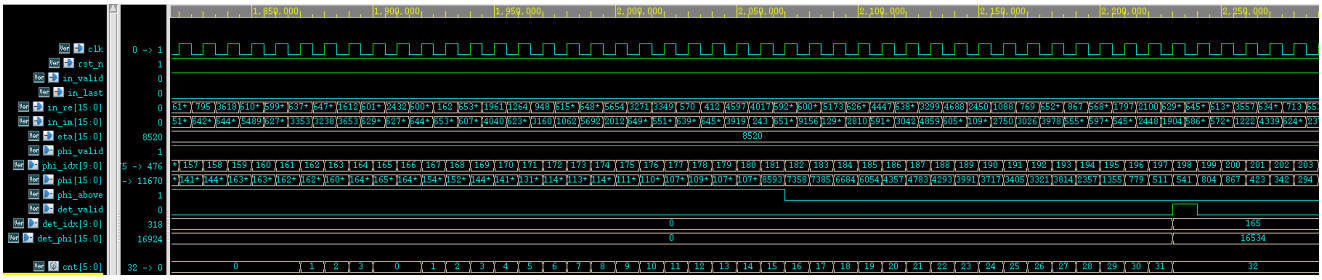
4.4.3 Behavioral Simulation

`q3.m` writes the quantized inputs, η and the bit-true golden data into `00_TESTBED/src`. `PATTERN.sv` feeds the 480 samples of each case, one per cycle, and compares every metric sample and every detection with the golden data. In Synopsys VCS S-2021.09, both the noiseless and the 12 dB case match the bit-true model for all 480 metric samples and all 3 detections.

The waveforms below are the noiseless case in Verdi, with a 10 ns clock. `cnt` is the counter inside `PEAK_SEARCH`. It is a register, so it shows the update one cycle after the metric sample that caused it. After a detection it stays at 32 until the next search starts.



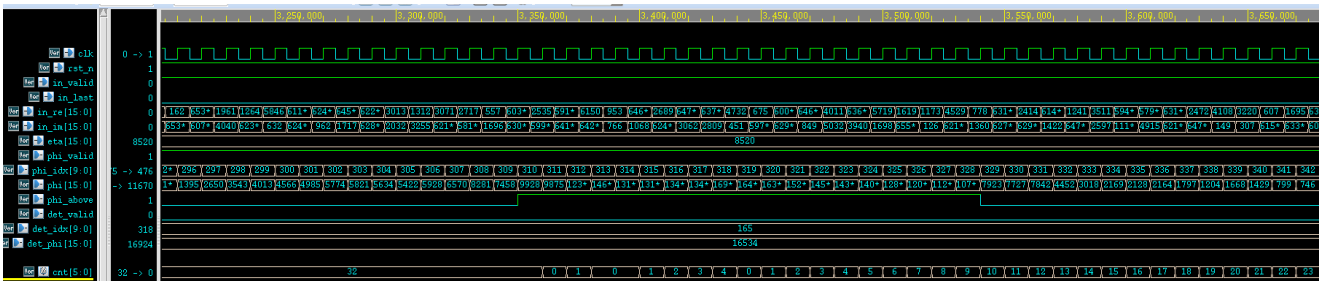
(a) `phi_above` rises at $m = 150$, where $\Phi = 8742 > \eta$.



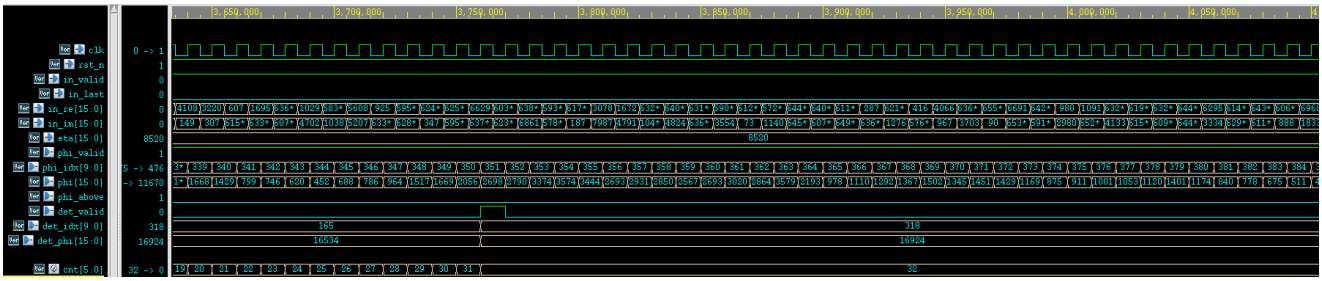
(b) `cnt` counts from 0 to 32 after the candidate $m = 165$; `det_valid` rises at `phi_idx` = 198.

Figure 4.3: Behavioral Simulation: First Boundary, $\hat{m} = 165$

In Figure 4.3, Φ crosses η at $m = 150$ and the search starts. Every larger value replaces the candidate and resets `cnt`; the last replacement is $m = 165$ with $\Phi = 16534$ (1.0092). From there `cnt` runs from 0 to 32 without another reset, `phi_above` falls at $m = 182$ ($\Phi = 7358$), and `det_valid` rises with `det_idx` = 165 and `det_phi` = 16534 while `phi_idx` is 198, i.e. $198 - 165 = 33$ cycles after the candidate.



(a) `phi_above` rises at $m = 310$; the candidate is replaced several times, last at $m = 318$.



(b) `cnt` reaches 32 and `det_valid` rises at `phi_idx` = 351.

Figure 4.4: Behavioral Simulation: Second Boundary, $\hat{m} = 318$

In Figure 4.4(a), the search starts at $m = 310$ ($\Phi = 9928$), and `cnt` returns to 0 each time a larger value arrives, last at $m = 318$ with $\Phi = 16924$ (1.0330). In (b), `cnt` reaches 32 and `det_valid` reports `det_idx` = 318 while `phi_idx` is 351, again $351 - 318 = 33$ cycles later.

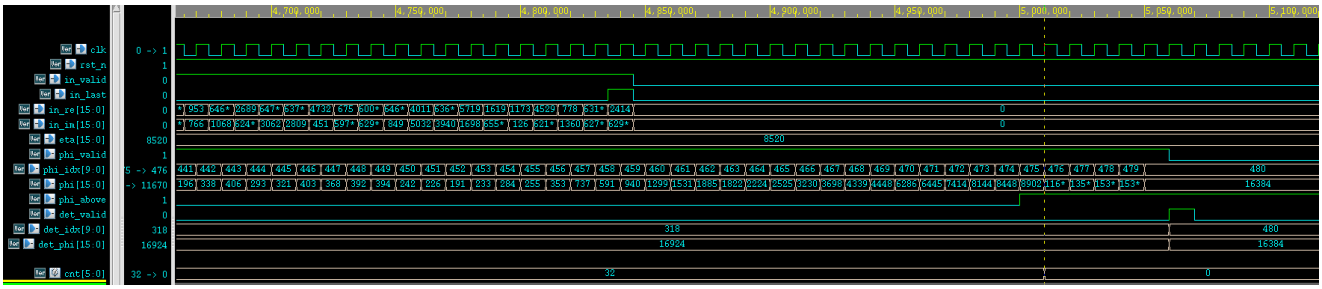


Figure 4.5: Behavioral Simulation: Last Boundary, $\hat{m} = 480$, Output by the Flush Rule

In Figure 4.5, `in_last` marks the last input sample and `in_valid` falls after it. The search for the third lobe starts at $m = 475$ ($\Phi = 8902$). The last metric sample, $m = 480$ with $\Phi = 16384$ (= 1.0), arrives 22 cycles after `in_last`, and the flush rule outputs it one cycle later (`det_idx` = 480, `det_phi` = 16384) instead of waiting for 32 more samples.

4.4.4 Comparison with the Simulation

Since the RTL output equals the bit-true model `sbd_model.m` bit for bit, Figure 4.6 draws the hardware metric from that model on top of the floating-point Φ_{NM} of Q2.

Table 4.3: Detected Boundaries: HDL versus MATLAB

	HDL	MATLAB float, same searcher	MATLAB, argmax every symbol (Q2)
Noiseless	165, 318, 480	165, 318, 480	160, 318, 480
SNR 12 dB	162, 320, 480	162, 320, 480	159, 162, 480

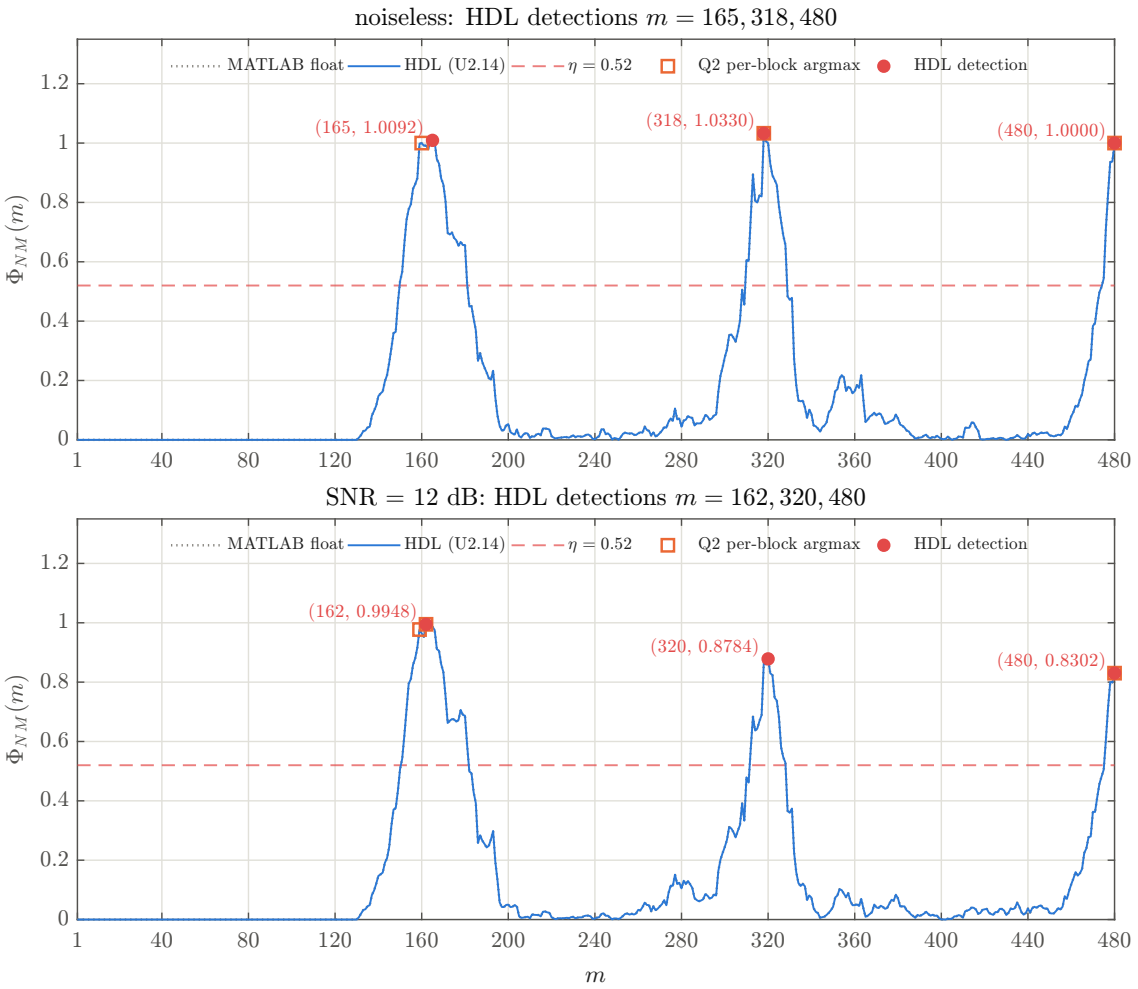


Figure 4.6: Hardware Metric from the Bit-True Model, Equal to the RTL Output (blue), and Floating-Point Φ_{NM} (dotted), with the Hardware Detections (dots) and the Argmax of Every Symbol from Q2 (squares)

The two metric curves overlap; they differ by at most 9.3×10^{-5}. Running the same peak searcher on the floating-point metric gives the same boundaries as the hardware, so quantization does not change any decision.

The differences from Q2 come from the decision rule. Q2 takes the maximum inside fixed 160-sample blocks, while the hardware takes the maximum of each lobe above η. At 12 dB this fixes symbol 2: the hardware finds 320, the correct symbol end, where the block rule picked 162 from the previous lobe. In the noiseless case, though, the hardware reports 165 for symbol 1 instead of 160, because $\Phi_{NM}(165) = 1.0092$ is 0.0092 larger than $\Phi_{NM}(160) = 1$, the overshoot explained in Q2(c). The block rule could not pick 165 because 165 lies in the next block.

AI Tool Disclosure

Please disclose how you use AI tools and its version.

I used **Claude Opus 5.5** to help with this homework in the following five parts:

(1) Understanding the homework specification.

(2) Assisting with the MATLAB code.

(3) Assisting with the SystemVerilog RTL and testbench.

(4) Plotting the figures.

(5) Writing this LaTeX report.